ASSIGNMENT – MACHINE LEARNING

Customer Churn Prediction and Segmentation System

Problem Statement

A service-based company is experiencing customer attrition and wants to use historical customer data to identify customers who are likely to discontinue its services. Develop a **machine learning system** that predicts customer churn and also groups customers with similar behavioural characteristics to support targeted retention strategies.

Technical Requirements

- Select an appropriate **public customer churn dataset** and clearly mention its source.
- Perform necessary **data preprocessing**, including:
  - handling missing values,
  - encoding categorical attributes,
  - removing irrelevant attributes, and
  - feature scaling wherever required.
- Perform suitable **exploratory data analysis** to understand important characteristics of the dataset.
- Divide the dataset appropriately into **training and testing data**.
- Implement at least **three supervised learning algorithms** from the syllabus, such as:
  - Logistic Regression,
  - Naive Bayes,
  - Decision Tree,
  - Support Vector Machine,
  - Neural Network,
  - Bagging, or
  - Boosting.
- Train the selected models to classify customers as **Churn / Non-Churn**.
- Evaluate the models using suitable performance measures such as:
  - Accuracy,
  - Precision,
  - Recall,
  - F1-score, and
  - Confusion Matrix.

- Compare the performance of the implemented models and identify the **best-performing model** with suitable justification.
- Apply **K-means clustering** to group customers based on relevant behavioural or service-related attributes.
- Determine an appropriate number of clusters and interpret the characteristics of the resulting customer groups.
- Apply **PCA** wherever appropriate to reduce dimensionality or support visualization of customer clusters.
- Provide meaningful visualizations for:
    - important dataset characteristics,
    - classification performance, and
    - customer clusters.
- Use the final results to identify **high-risk customer groups** and provide suitable retention recommendations.
- The implementation should be executable from beginning to end without errors.

Expected Outcome

The completed system should demonstrate a complete machine-learning workflow from **data preprocessing → model training → prediction → model evaluation and selection → customer clustering → interpretation of results**.

The selected concepts directly reflect the course emphasis on supervised and unsupervised learning and the course outcome concerning evaluation and model selection.

Assignment Submission Instructions

- This is a **team assignment**.
- Only **one submission per team** is required.
- Submit all components in **one consolidated document/file**.
- The submitted document must include:
    - **Pseudocode / workflow** of the proposed machine learning solution.
    - **Dataset description and source**.
    - **Data preprocessing steps**.
    - **Implementation/code**.
    - **Screenshots of important outputs and visualizations**.
    - **Performance comparison of the implemented models**.
    - **Final observations and conclusion**.
    - **One-page individual contribution write-up from every team member**.

- Individual contributions may include:
    - dataset preparation,
    - preprocessing,
    - exploratory analysis,
    - model implementation,
    - hyperparameter experimentation,
    - evaluation,
    - clustering,
    - visualization,
    - testing,
    - documentation, or
    - GitHub management.
- **Do not mention student names or register numbers anywhere in the submitted document.**
- Plagiarism checking will be enabled.
- The plagiarism/similarity percentage must be **below 20%**.
- Submissions with **20% or more similarity must be revised and resubmitted**.
- Upload the **complete working code/project to GitHub**.
- The GitHub repository must contain the dataset link/reference, source files, required libraries/dependencies, and all files needed to execute the project.
- Ensure that the submitted implementation is completely tested before final submission.

INTELLIGENT CUSTOMER CHURN RISK PREDICTION AND BEHAVIOURAL SEGMENTATION SYSTEM

ABSTRACT

Customer churn is a major challenge for service-based organizations because losing existing customers can negatively affect revenue, customer relationships, and long-term business growth. Identifying customers who are likely to discontinue a service at an early stage allows organizations to take appropriate retention measures.

This project presents an end-to-end machine learning system for customer churn prediction and behavioural segmentation. A publicly available customer churn dataset is used to analyze customer demographics, service subscriptions, account information, contract characteristics, payment methods, tenure, and billing-related attributes.

The proposed system performs data preprocessing, exploratory data analysis, feature transformation, supervised classification, model evaluation, customer segmentation, dimensionality reduction, and retention-oriented interpretation. Multiple supervised machine learning algorithms are implemented and compared using accuracy, precision, recall, F1-score, and confusion matrices. K-Means clustering is subsequently applied to identify groups of customers with similar behavioural and service characteristics. Principal Component Analysis (PCA) is used to support visualization of the resulting customer segments.

The final system combines churn prediction and customer segmentation to identify high-risk customer groups and associate them with suitable retention strategies. Thus, the proposed solution moves beyond simple churn prediction and provides a decision-support framework for understanding who is likely to churn, what type of customer they are, and what type of intervention may be appropriate.

INTRODUCTION

Customer retention is an important objective for service-oriented organizations. Customers may discontinue their services because of pricing, contract conditions, poor service experience, lack of technical support, changing requirements, or alternative service providers.

Traditional customer management approaches generally treat customers in a similar manner. However, customers have different characteristics, usage patterns, service preferences, contract types, and levels of engagement. Machine learning can help organizations identify hidden patterns within historical customer data and use those patterns to make informed decisions.

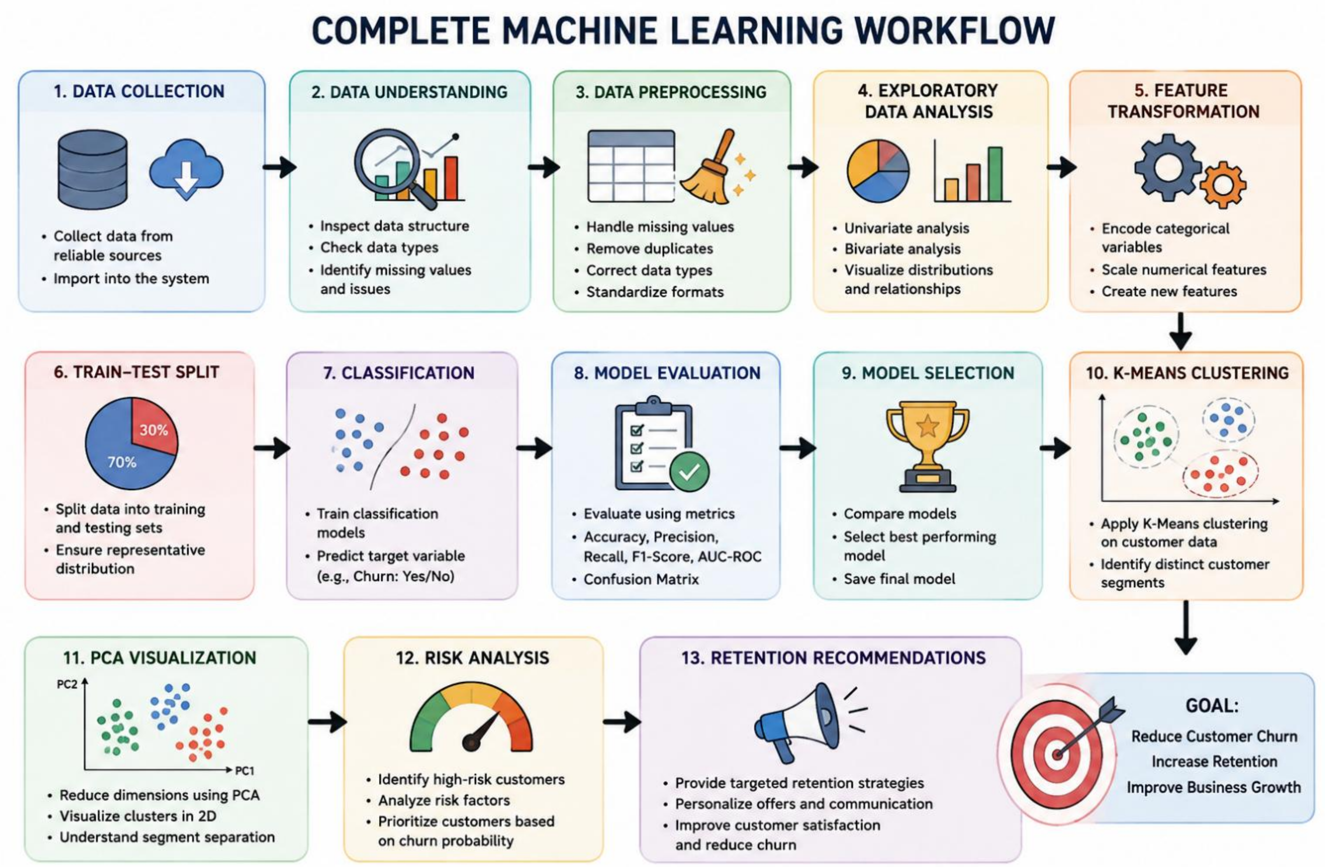
The proposed project addresses this problem using two complementary machine learning approaches.

Supervised learning is used to predict whether a customer is likely to churn or remain with the service.

Unsupervised learning is used to group customers according to similarities in their behavioural and service-related characteristics.

The combination of these approaches creates a more informative customer intelligence system. Instead of producing only a binary churn prediction, the system can identify customer segments and analyze the churn behaviour of each segment.

The project follows a complete machine learning workflow:



PROBLEM STATEMENT

A service-based company is experiencing customer attrition and wants to use historical customer information to identify customers who are likely to discontinue its services. The objective is to develop a machine learning system that predicts customer churn and groups customers with similar behavioural characteristics so that targeted retention strategies can be developed.

OBJECTIVES

The major objectives of the proposed system are:

- To analyze historical customer information and identify factors associated with churn.
- To preprocess customer data and prepare it for machine learning.
- To perform exploratory data analysis to understand customer behaviour.
- To implement multiple supervised machine learning algorithms for churn classification.
- To evaluate and compare the implemented models using suitable performance metrics.
- To select the most appropriate model based on the evaluation results.
- To apply K-Means clustering to identify behavioural customer segments.
- To determine an appropriate number of clusters using clustering evaluation techniques.

- To apply PCA for dimensionality reduction and cluster visualization where appropriate.
- To identify high-risk customer segments.
- To provide suitable retention recommendations based on customer risk and behavioural characteristics.
- To develop a complete, executable implementation that can be tested from beginning to end.

PROPOSED SYSTEM

The proposed system consists of two major machine learning components.

Churn Prediction Component

The supervised learning component treats churn as a binary classification problem.

The target variable is:

Churn

where:

- **1 / Yes** represents a customer who churned.
- **0 / No** represents a customer who did not churn.

Multiple classification algorithms are trained using historical customer information. Their predictions are evaluated and compared to determine the most suitable model.

Customer Segmentation Component

The unsupervised learning component uses K-Means clustering to group customers with similar characteristics.

Unlike classification, clustering does not require a predefined target class. Instead, it discovers natural groups within the customer population based on selected behavioural and service-related attributes.

Integrated Decision-Support Component

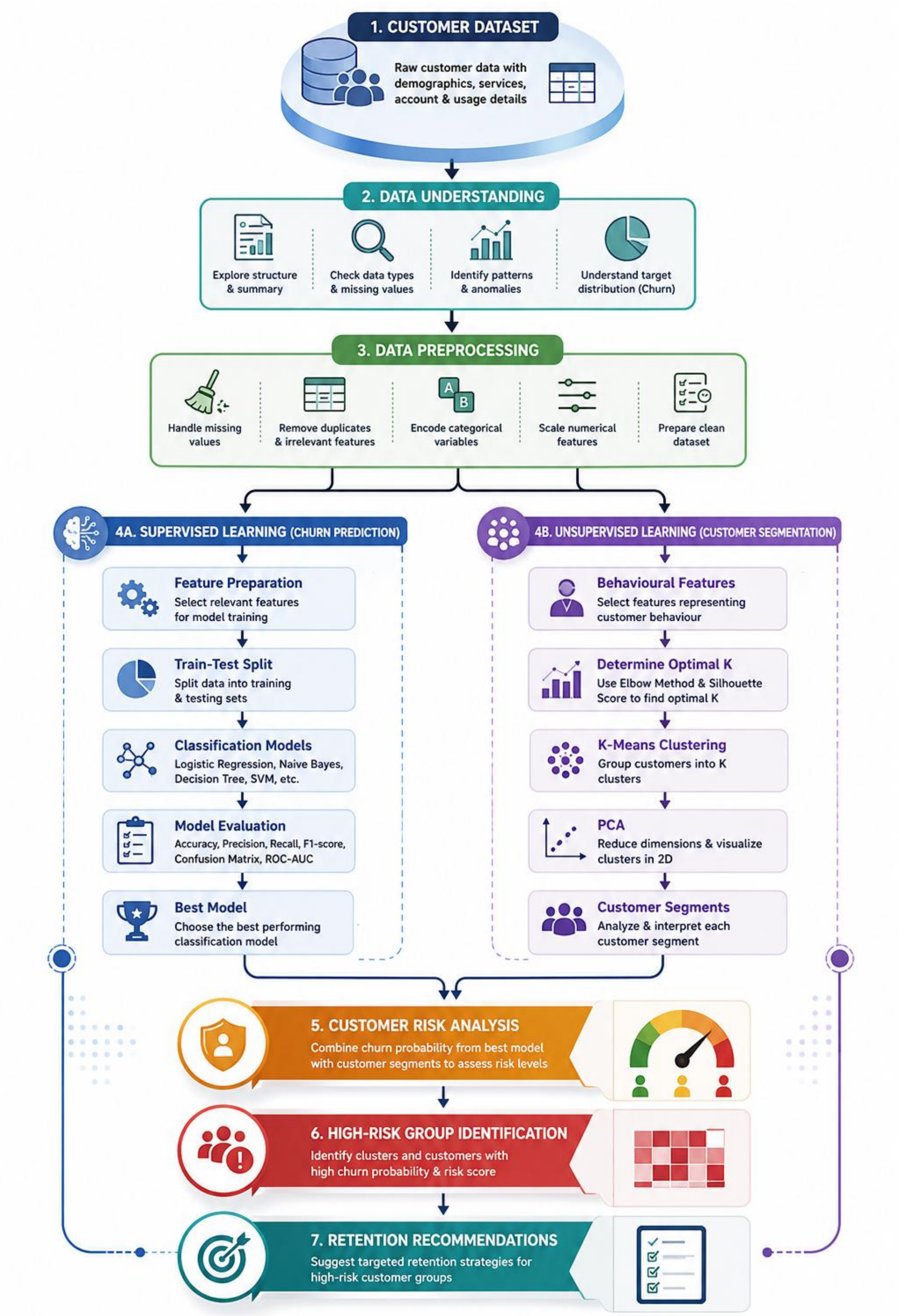
The final system combines:

Churn Probability + Customer Segment + Segment Churn Rate

to identify customers and groups requiring greater retention attention.

This provides a more practical interpretation of machine learning results.

SYSTEM WORKFLOW



PSEUDOCODE

BEGIN

Load the customer churn dataset

Inspect dataset dimensions, attributes and data types

Identify missing, inconsistent and duplicate values

Remove irrelevant attributes

Convert numerical attributes into appropriate numeric format

Handle missing values

Encode categorical attributes

Separate input features and churn target

Apply feature scaling where required

Split the dataset into training and testing sets

Initialize selected supervised learning algorithms

FOR each classification model

    Train the model using the training dataset

    Generate churn predictions for test data

    Calculate accuracy

    Calculate precision

    Calculate recall

    Calculate F1-score

    Generate confusion matrix

END FOR

Compare classification models

Select the most suitable model

Use the selected model to estimate customer churn probability

Prepare relevant behavioural attributes for clustering

Test different numbers of clusters

Apply Elbow Method

Calculate Silhouette Scores

Select an appropriate number of clusters

Apply K-Means clustering

Calculate characteristics of each cluster

Calculate churn rate for each cluster

Apply PCA when required for dimensionality reduction

Generate cluster visualization

Identify high-risk customer segments

Combine churn risk and cluster characteristics

Generate suitable retention recommendations

Display final results and visualizations

END

DATASET DESCRIPTION

Dataset Link: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Dataset: Telco Customer Churn
Source: Kaggle
Original Source: IBM Sample Data Sets
Records: 7,043 customers
Attributes: 21 columns
Target Variable: Churn

The project uses a publicly available Telco Customer Churn dataset.

The dataset contains historical customer-level information associated with telecommunications services. It includes customer demographic information, subscribed services, account information, contract details, billing information, payment methods, and churn status.

Important attributes include:

- Customer identifier
- Gender
- Senior citizen status
- Partner status
- Dependents
- Tenure
- Phone service
- Multiple lines
- Internet service
- Online security
- Online backup
- Device protection
- Technical support
- Streaming services
- Contract type
- Paperless billing

- Payment method
- Monthly charges
- Total charges
- Churn status

Target Variable

The target variable is **Churn**.

It represents whether the customer discontinued the service.

The target is transformed into a binary representation suitable for classification algorithms.

Dataset Source

The dataset source and access reference should be clearly mentioned in the final document and in the GitHub repository. The assignment specifically requires a public customer churn dataset with its source clearly identified.

```
CUSTOMER CHURN PREDICTION & BEHAVIOURAL SEGMENTATION SYSTEM
===========================================================================

All libraries imported successfully.

Output directories created.

===========================================================================
UPLOAD KAGGLE DATASET
===========================================================================

Please upload:
WA_Fn-UseC_-Telco-Customer-Churn.csv

Choose Files WA_Fn-Us...er-Churn.csv
WA_Fn-UseC_-Telco-Customer-Churn.csv(text/csv) - 977501 bytes, last modified: 9/2/2026 - 100% done
Saving WA_Fn-UseC_-Telco-Customer-Churn.csv to WA_Fn-UseC_-Telco-Customer-Churn (1).csv

Dataset loaded successfully.
File: WA_Fn-UseC_-Telco-Customer-Churn (1).csv
Rows: 7043
Columns: 21

First five records:
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	Pa
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	

5 rows × 21 columns

```
===========================================================================
DATA UNDERSTANDING
===========================================================================

Dataset Shape:
(7043, 21)

Column Names:
1 . customerID
2 . gender
3 . SeniorCitizen
4 . Partner
5 . Dependents
6 . tenure
7 . PhoneService
8 . MultipleLines
9 . InternetService
10 . OnlineSecurity
11 . OnlineBackup
12 . DeviceProtection
13 . TechSupport
14 . StreamingTV
15 . StreamingMovies
16 . Contract
17 . PaperlessBilling
18 . PaymentMethod
19 . MonthlyCharges
20 . TotalCharges
21 . Churn
```

	Data Type
customerID	object
gender	object
SeniorCitizen	int64
Partner	object
Dependents	object
tenure	int64
PhoneService	object
MultipleLines	object
InternetService	object
OnlineSecurity	object
OnlineBackup	object
DeviceProtection	object
TechSupport	object
StreamingTV	object
StreamingMovies	object
Contract	object
PaperlessBilling	object
PaymentMethod	object
MonthlyCharges	float64
TotalCharges	object
Churn	object

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

Statistical Summary:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
customerID	7043	7043	3186-AJIEK	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
gender	7043	2	Male	3555	NaN	NaN	NaN	NaN	NaN	NaN	NaN
SeniorCitizen	7043.0	NaN	NaN	NaN	0.162147	0.368612	0.0	0.0	0.0	0.0	1.0
Partner	7043	2	No	3641	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Dependents	7043	2	No	4933	NaN	NaN	NaN	NaN	NaN	NaN	NaN
tenure	7043.0	NaN	NaN	NaN	32.371149	24.559481	0.0	9.0	29.0	55.0	72.0
PhoneService	7043	2	Yes	6361	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MultipleLines	7043	3	No	3390	NaN	NaN	NaN	NaN	NaN	NaN	NaN
InternetService	7043	3	Fiber optic	3096	NaN	NaN	NaN	NaN	NaN	NaN	NaN
OnlineSecurity	7043	3	No	3498	NaN	NaN	NaN	NaN	NaN	NaN	NaN
OnlineBackup	7043	3	No	3088	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DeviceProtection	7043	3	No	3095	NaN	NaN	NaN	NaN	NaN	NaN	NaN
TechSupport	7043	3	No	3473	NaN	NaN	NaN	NaN	NaN	NaN	NaN
StreamingTV	7043	3	No	2810	NaN	NaN	NaN	NaN	NaN	NaN	NaN
StreamingMovies	7043	3	No	2785	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Contract	7043	3	Month-to-month	3875	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PaperlessBilling	7043	2	Yes	4171	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PaymentMethod	7043	4	Electronic check	2365	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MonthlyCharges	7043.0	NaN	NaN	NaN	64.761692	30.090047	18.25	35.5	70.35	89.85	118.75
TotalCharges	7043	6531		11	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Churn	7043	2	No	5174	NaN	NaN	NaN	NaN	NaN	NaN	NaN

DATA PREPROCESSING

Raw customer data cannot be directly supplied to all machine learning algorithms. Therefore, several preprocessing operations are performed.

Data Inspection

The dataset is first examined to determine:

- Number of records
- Number of attributes
- Data types
- Missing values
- Duplicate records
- Unique values
- Distribution of the target variable

This provides an understanding of the quality and structure of the dataset.

Removal of Irrelevant Attributes

The customer identification attribute is removed because it uniquely identifies a customer but does not represent meaningful behavioural information.

For example:

customerID

is not used as a predictive feature.

Missing Value Handling

Missing values are identified using null-value analysis.

Some numerical fields may contain blank values rather than conventional missing-value indicators. Such fields are converted into appropriate numeric types and missing observations are handled using a suitable strategy.

Data Type Conversion

Attributes representing numerical quantities are converted into numeric data types.

For example:

TotalCharges

is converted from text representation into a numerical format.

Categorical Encoding

Machine learning algorithms require numerical representations of categorical attributes.

Binary attributes are transformed into numerical values where appropriate.

Multi-category variables such as contract type, internet service and payment method are transformed using one-hot encoding.

Feature Scaling

Numerical features are standardized using a scaling technique such as StandardScaler.

Scaling is particularly important for algorithms such as:

- Support Vector Machine
- K-Means
- PCA

because these methods can be influenced by differences in feature magnitude.

Train-Test Split

The preprocessed dataset is divided into training and testing subsets.

A stratified split can be used so that the proportion of churn and non-churn customers remains approximately consistent across both subsets.

```
Missing Values:
```

	Missing Values	Missing Percentage
customerID	0	0.0
gender	0	0.0
SeniorCitizen	0	0.0
Partner	0	0.0
Dependents	0	0.0
tenure	0	0.0
PhoneService	0	0.0
MultipleLines	0	0.0
InternetService	0	0.0
OnlineSecurity	0	0.0
OnlineBackup	0	0.0
DeviceProtection	0	0.0
TechSupport	0	0.0
StreamingTV	0	0.0
StreamingMovies	0	0.0
Contract	0	0.0
PaperlessBilling	0	0.0
PaymentMethod	0	0.0
MonthlyCharges	0	0.0
TotalCharges	0	0.0
Churn	0	0.0

```
Duplicate Rows: 0

Target Distribution:
```

	Count
Churn	
No	5174
Yes	1869

```
Target Percentage:
```

	Percentage
Churn	
No	73.46
Yes	26.54

```
============================================================
DATA CLEANING
============================================================
Duplicate rows removed: 0

Missing TotalCharges after conversion: 11

Final cleaned shape: (7043, 21)
```

```
Remaining missing values:
```

	Missing Values
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis is performed before model development to understand relationships within the dataset and identify characteristics associated with customer churn.

The following visualizations should be included in the final report.

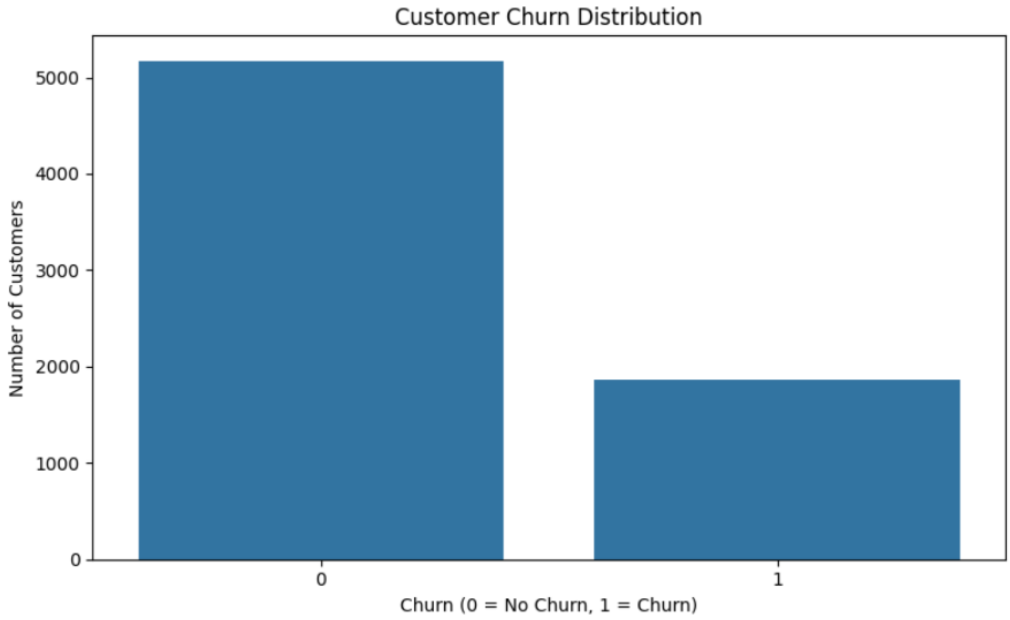
Churn Distribution

A count plot showing the number of:

- Churned customers
- Non-churned customers

This helps determine the overall churn distribution and whether class imbalance is present.

```
...  ==========================================================================
     EXPLORATORY DATA ANALYSIS
     ==========================================================================
```

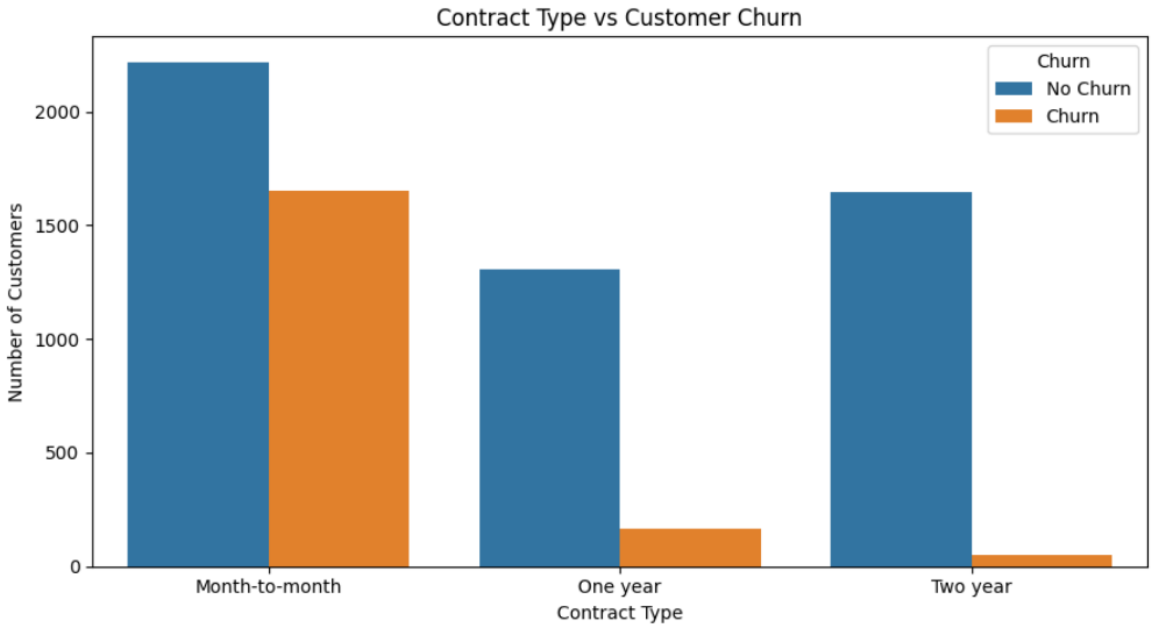



```
Overall Customer Churn Rate: 26.54%

No Churn: 73.46 %
Churn: 26.54 %
```

Churn by Contract Type

Customers are grouped according to contract type and their churn rates are compared.

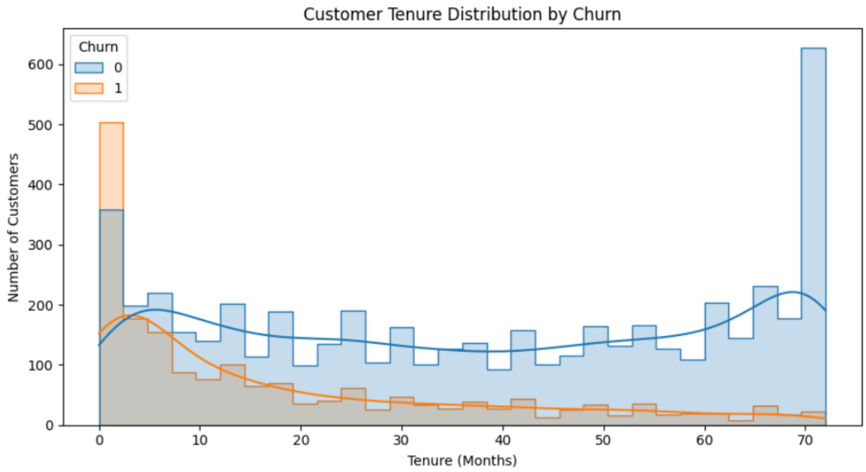
This helps determine whether contractual commitment is associated with customer retention.

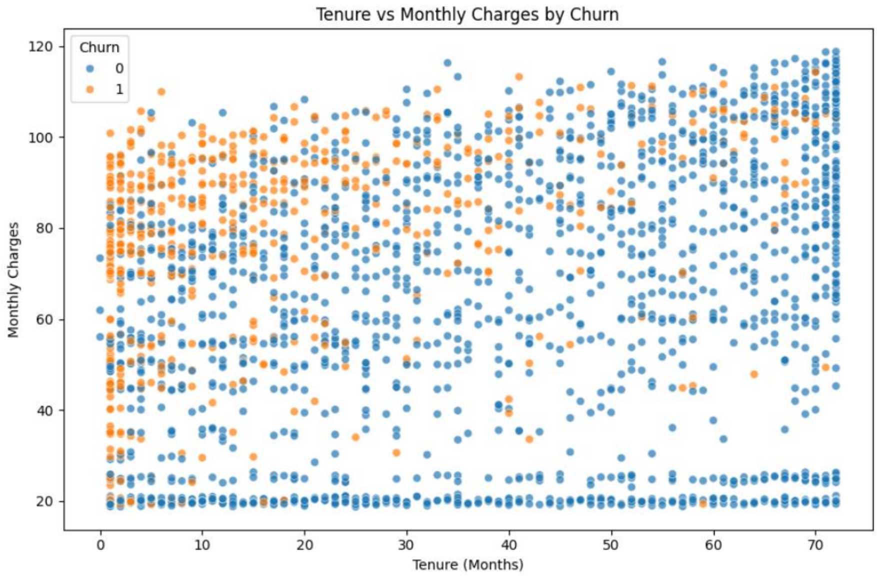


Tenure Distribution

Tenure is analyzed with respect to churn.

The analysis helps identify whether customers with shorter or longer relationships with the company demonstrate different churn patterns.

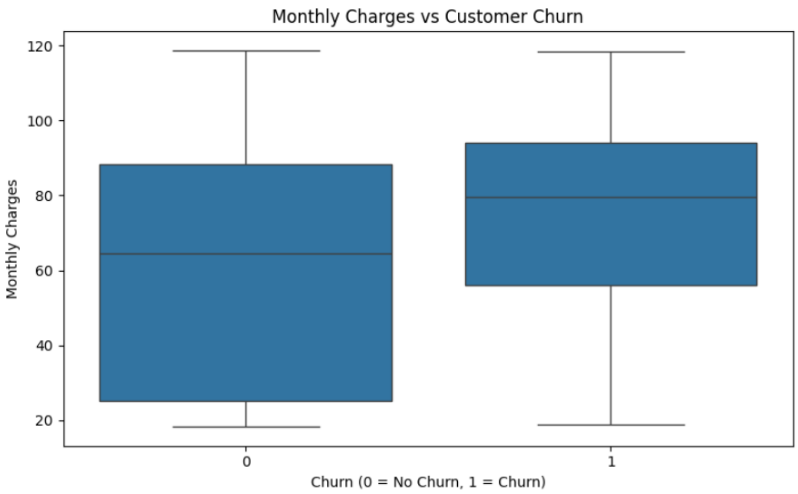




Monthly Charges and Churn

Monthly charges are compared between churned and non-churned customers.

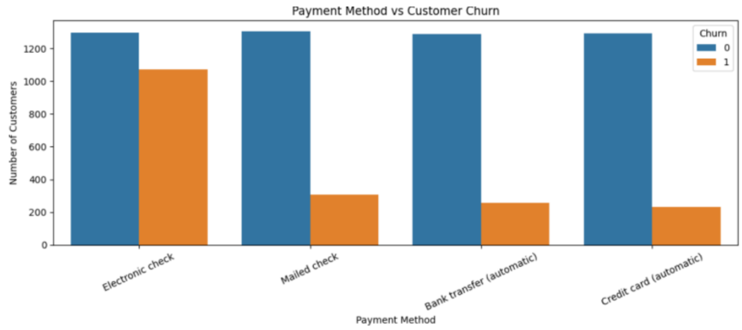
This can help determine whether pricing or billing level may be associated with customer attrition.



Payment Method and Churn

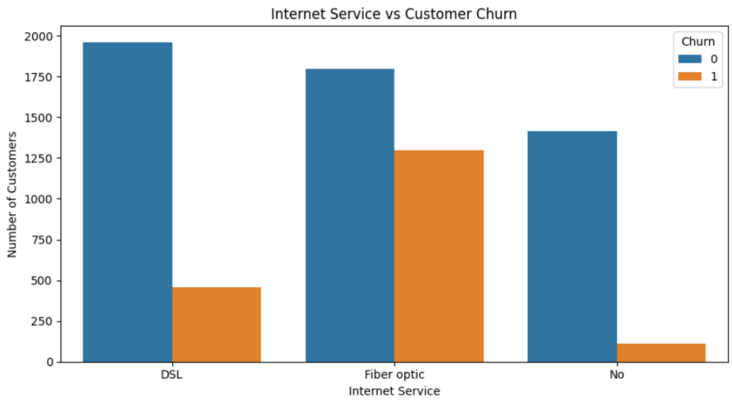
Customer churn is analyzed across different payment methods.

This may reveal behavioural differences among customer groups.



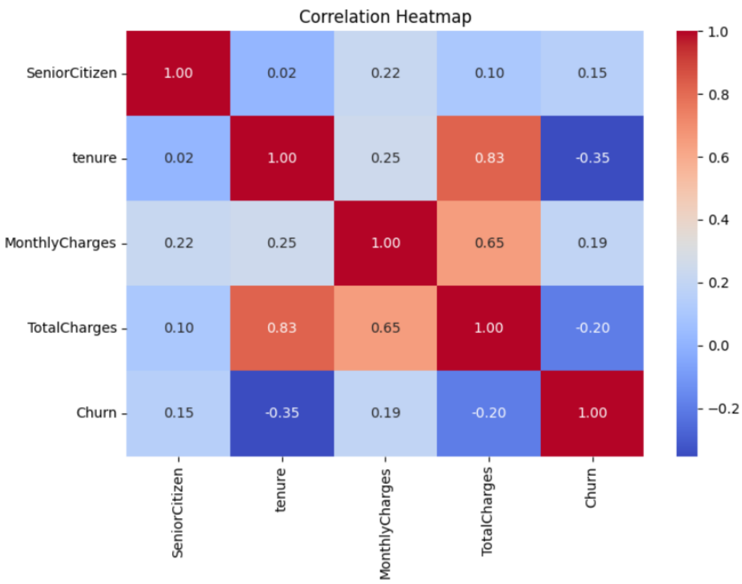
Service Subscription and Churn

Internet service, technical support, online security and related services can be analyzed to identify service-related patterns associated with churn.



Correlation Analysis

A correlation heatmap can be generated for suitable numerical attributes to identify relationships among important variables.



MACHINE LEARNING MODELS

The assignment requires at least three supervised learning algorithms.

The proposed implementation can use the following models.

Logistic Regression

Logistic Regression is used as an interpretable baseline classification algorithm.

It estimates the probability that a customer belongs to the churn class.

The model is particularly useful because it provides probability-based predictions and allows the effect of input features to be analyzed.

Naive Bayes

Naive Bayes is a probabilistic classification technique based on Bayes' theorem.

It assumes conditional independence between features given the target class.

It provides a computationally efficient comparison against other classification approaches.

Decision Tree

Decision Tree classification uses a sequence of feature-based decisions to divide customers into different classes.

The model is easy to interpret because its decisions can be represented as a tree structure.

Support Vector Machine

Support Vector Machine attempts to identify an optimal decision boundary separating churn and non-churn customers.

Because SVM is sensitive to feature scale, standardized input features are used.

MODEL TRAINING

The dataset is divided into:

Training Data

Used to learn patterns from historical customer records.

Testing Data

Used to evaluate how well the trained model performs on previously unseen customer records.

The models are trained using the same training and testing framework to ensure a fair comparison.

```
...
==========================================================================
FEATURE AND TARGET PREPARATION
==========================================================================

Numerical Features:
['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']

Categorical Features:
['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',

Total Numerical Features: 4
Total Categorical Features: 15

==========================================================================
FEATURE TRANSFORMATION AND PREPROCESSING
==========================================================================
Preprocessing pipeline created successfully.

==========================================================================
TRAIN-TEST SPLIT
==========================================================================
Training Samples: 5634
Testing Samples: 1409

Training Churn Distribution:
```

Churn	Percentage
0	73.46
1	26.54

```
Testing Churn Distribution:
```

Churn	Percentage
0	73.46
1	26.54

```
...
==========================================================================
CLASSIFICATION MODELS
==========================================================================
✓ Logistic Regression
✓ Naive Bayes
✓ Decision Tree
✓ Support Vector Machine

==========================================================================
MODEL TRAINING
==========================================================================

Training Logistic Regression...
✓ Logistic Regression training completed.

Training Naive Bayes...
✓ Naive Bayes training completed.

Training Decision Tree...
✓ Decision Tree training completed.

Training Support Vector Machine...
✓ Support Vector Machine training completed.

All classification models trained successfully.
```

HYPERPARAMETER EXPERIMENTATION

Selected model parameters can be varied to study their influence on classification performance.

For example, Decision Tree experiments can compare different values of maximum tree depth.

SVM experiments can compare different values of the regularization parameter **C**.

Logistic Regression can also be evaluated using different regularization strengths.

The final parameters should be selected based on the actual experimental results obtained during implementation.

A table such as the following can be included:

Model	**Parameter**	**Values Tested**	**Selected Value**
Logistic Regression	C	0.1, 1, 10	Actual result
Decision Tree	max_depth	3, 5, 8, 10	Actual result
SVM	C	0.1, 1, 10	Actual result

```
===========================================================================
HYPERPARAMETER EXPERIMENTATION
===========================================================================
Running Decision Tree hyperparameter search...

Best Parameters:
{'classifier__max_depth': 5, 'classifier__min_samples_split': 2}

Best Cross-Validation F1-Score: 0.56

Tuned Decision Tree Test Performance:
Accuracy: 0.7984
Precision: 0.6347
Recall: 0.5668
F1-Score: 0.5989
ROC-AUC: 0.8297
```

MODEL EVALUATION

The implemented classification models are evaluated using multiple performance measures rather than relying only on accuracy.

Accuracy

Accuracy represents the proportion of correctly classified customers among all tested customers.

Precision

Precision measures how many customers predicted as churners were actually churners.

Recall

Recall measures how many of the actual churners were successfully identified by the model.

Recall is particularly important in a churn prediction problem because failing to identify an actual churner may result in a missed retention opportunity.

F1-Score

F1-score combines precision and recall into a single metric and is useful when both types of performance need to be considered.

Confusion Matrix

The confusion matrix provides the number of:

- True Positives
- True Negatives
- False Positives
- False Negatives

for each model.

```
==========================================================================
MODEL EVALUATION
==========================================================================

Model Performance Comparison:
```

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.8055	0.6572	0.5588	0.6040	0.8419
1	Naive Bayes	0.6948	0.4589	0.8369	0.5928	0.8074
2	Support Vector Machine	0.7913	0.6418	0.4840	0.5518	0.7905
3	Decision Tree	0.7289	0.4896	0.5053	0.4974	0.6573

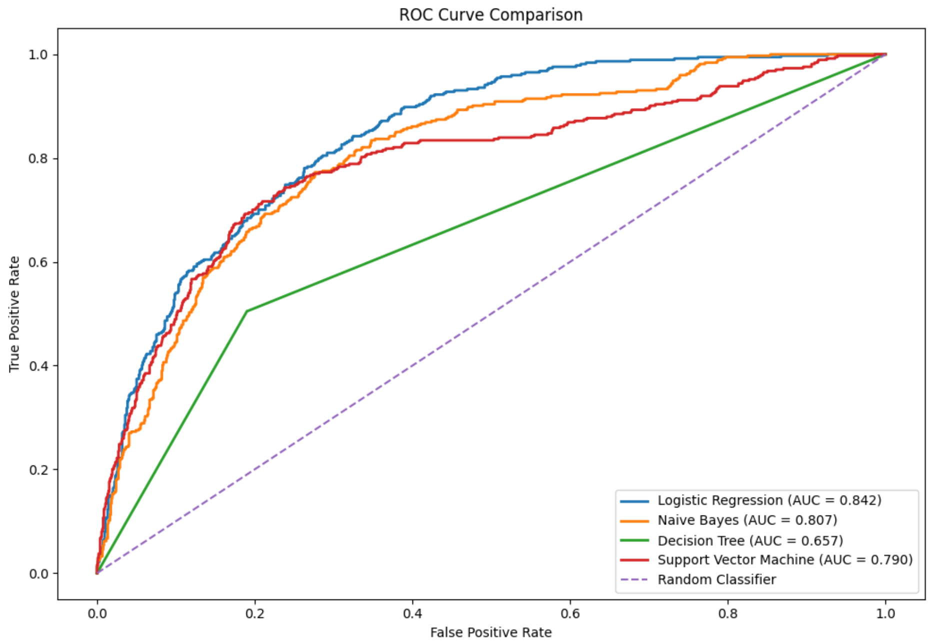
```
==========================================================================
CLASSIFICATION REPORTS
==========================================================================

----------------------------------------------------------------
Logistic Regression
----------------------------------------------------------------
              precision    recall  f1-score   support

   Non-Churn       0.85      0.89      0.87      1035
       Churn       0.66      0.56      0.60       374

    accuracy                           0.81      1409
   macro avg       0.75      0.73      0.74      1409
weighted avg       0.80      0.81      0.80      1409


----------------------------------------------------------------
Naive Bayes
----------------------------------------------------------------
              precision    recall  f1-score   support

   Non-Churn       0.92      0.64      0.76      1035
       Churn       0.46      0.84      0.59       374

    accuracy                           0.69      1409
   macro avg       0.69      0.74      0.67      1409
weighted avg       0.79      0.69      0.71      1409
```

```
...
-----------------------------------------------------------------
Decision Tree
-----------------------------------------------------------------
              precision    recall  f1-score   support

   Non-Churn       0.82      0.81      0.81      1035
       Churn       0.49      0.51      0.50       374

    accuracy                           0.73      1409
   macro avg       0.65      0.66      0.66      1409
weighted avg       0.73      0.73      0.73      1409


-----------------------------------------------------------------
Support Vector Machine
-----------------------------------------------------------------
              precision    recall  f1-score   support

   Non-Churn       0.83      0.90      0.86      1035
       Churn       0.64      0.48      0.55       374

    accuracy                           0.79      1409
   macro avg       0.74      0.69      0.71      1409
weighted avg       0.78      0.79      0.78      1409
```

ROC-AUC

If implemented, ROC-AUC can additionally be used to evaluate how effectively a model separates churners from non-churners across different classification thresholds.

```
======================================================================
ROC CURVE ANALYSIS
...
======================================================================
```



PERFORMANCE COMPARISON OF IMPLEMENTED MODELS

The final results should be presented in a comparison table generated from the actual implementation.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	Actual	Actual	Actual	Actual	Actual
Naive Bayes	Actual	Actual	Actual	Actual	Actual

Decision Tree	Actual	Actual	Actual	Actual	Actual
SVM	Actual	Actual	Actual	Actual	Actual

A comparative bar chart should also be included.

The best-performing model should be selected based on the project objective rather than simply choosing the highest accuracy.

For customer churn prediction, recall and F1-score for the churn class should receive particular consideration because identifying actual churners is an important objective.

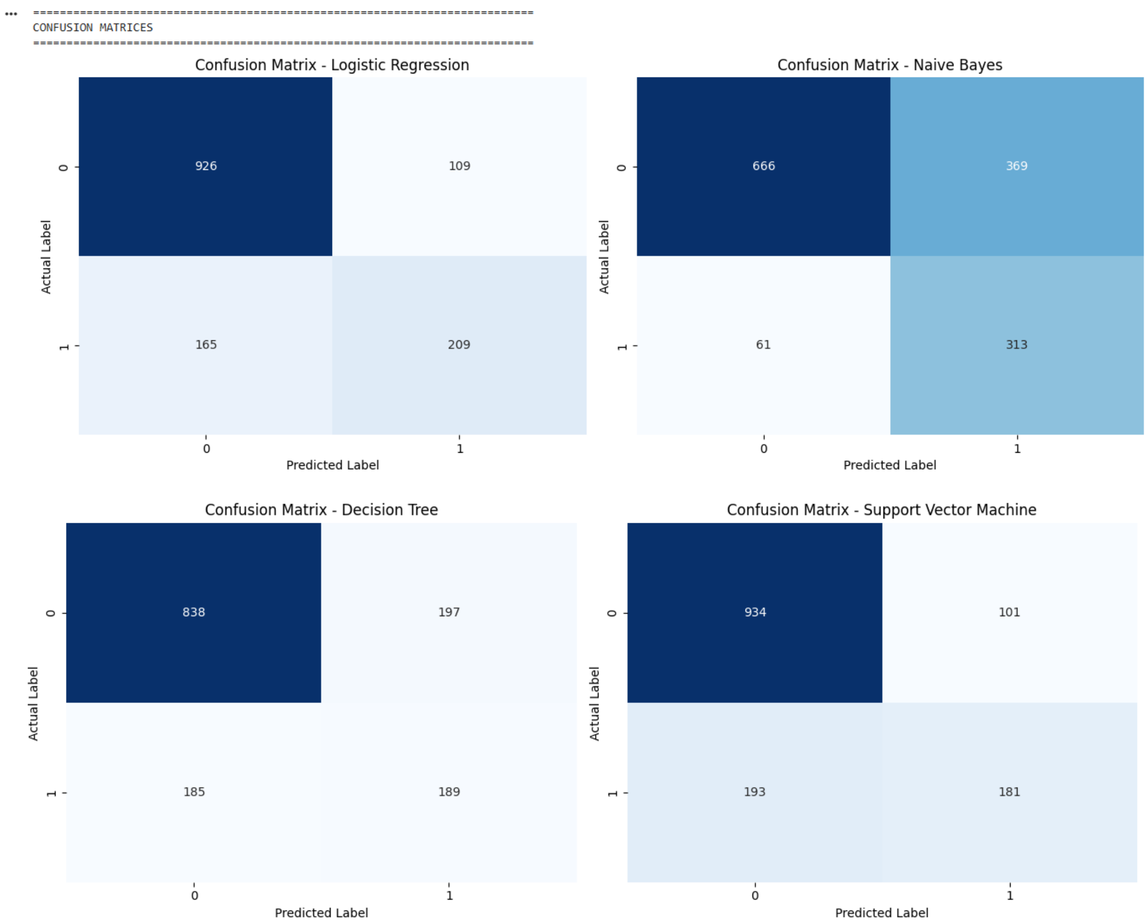
CONFUSION MATRIX ANALYSIS

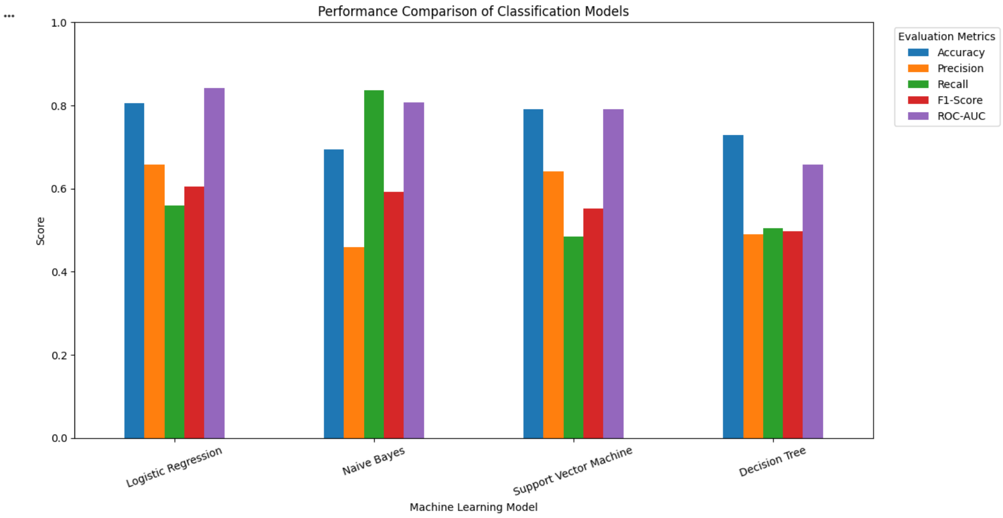
A confusion matrix is generated for each classification model.

The matrix consists of:

	Predicted Non-Churn	**Predicted Churn**
Actual Non-Churn	True Negative	False Positive
Actual Churn	False Negative	True Positive

A lower number of false negatives is desirable in a churn prediction application because false negatives represent customers who actually churned but were not identified by the system.





CUSTOMER CHURN RISK SCORE

An additional decision-support layer can be introduced after selecting the best-performing model.

Instead of providing only a binary prediction, the system can use the predicted churn probability to categorize customers into risk levels.

Example:

Churn Probability	Risk Level
Below 0.30	Low
0.30–0.60	Medium
Above 0.60	High

The thresholds should be treated as configurable values and can be adjusted according to the actual project requirements.

This allows the system to produce results such as:

Customer	Churn Probability	Risk
Customer A	Actual	High
Customer B	Actual	Medium
Customer C	Actual	Low

```
===========================================================================
BEST MODEL SELECTION
===========================================================================
Selected Best Model: Logistic Regression

Performance of Selected Model:
Accuracy  : 0.8055
Precision : 0.6572
Recall    : 0.5588
F1-Score  : 0.6040
ROC-AUC   : 0.8419
```

K-MEANS CUSTOMER SEGMENTATION

K-Means clustering is used to divide customers into groups based on similarities in their selected behavioural and service-related characteristics.

The clustering process does not use the churn label as the clustering target. Instead, customer attributes are used to discover naturally occurring customer groups.

Potential clustering attributes include:

- Tenure
- Monthly charges
- Total charges
- Service subscriptions
- Contract characteristics
- Other suitable numerical representations of customer behaviour

```
================================================================================
UNSUPERVISED LEARNING - K-MEANS
================================================================================
Clustering matrix shape: (7043, 45)
Clustering features: 19
```

DETERMINING THE NUMBER OF CLUSTERS

An appropriate value of K is determined before applying the final K-Means model.

Elbow Method

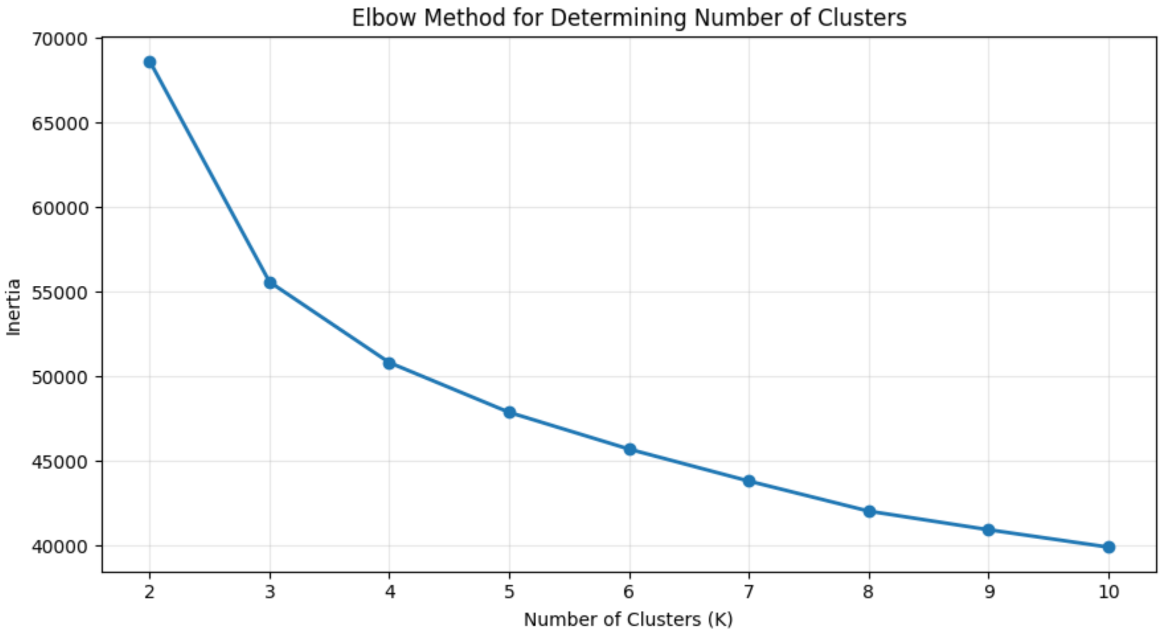
K-Means models are trained using different values of K.

The within-cluster sum of squares is plotted against the number of clusters.

The point at which the reduction in within-cluster variation begins to slow down can be considered an appropriate candidate for K.

```
================================================================================
ELBOW METHOD
================================================================================
```

	K	Inertia
0	2	68651.081466
1	3	55588.452564
2	4	50815.305920
3	5	47872.956220
4	6	45698.284591
5	7	43807.584480
6	8	42030.120958
7	9	40927.760400
8	10	39901.289772



Silhouette Score

Silhouette scores are calculated for different cluster counts.

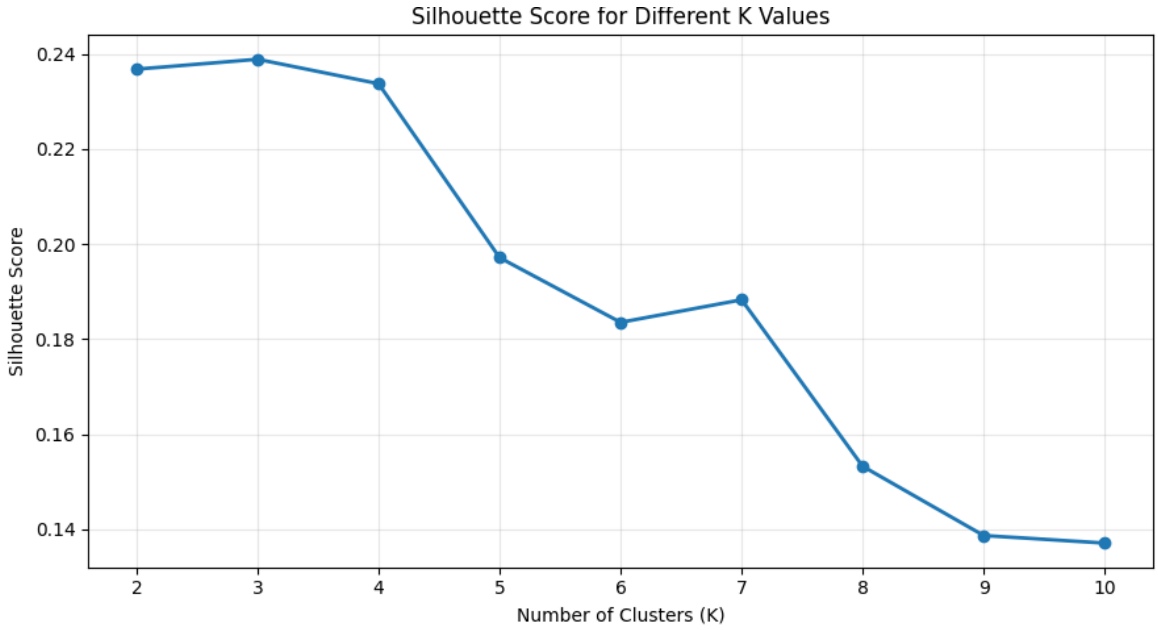
A higher score generally indicates better separation and cohesion of clusters.

The final cluster count should be selected by considering:

- Elbow Method
- Silhouette Score
- Cluster interpretability

```
==========================================================================
SILHOUETTE SCORE ANALYSIS
==========================================================================
```

	K	Silhouette Score
0	2	0.2368
1	3	0.2388
2	4	0.2337
3	5	0.1971
4	6	0.1835
5	7	0.1883
6	8	0.1533
7	9	0.1387
8	10	0.1371




```
===========================================================================
OPTIMAL NUMBER OF CLUSTERS
===========================================================================
Selected Optimal K: 3
Best Silhouette Score: 0.2388

===========================================================================
FINAL K-MEANS CLUSTERING
===========================================================================
K-Means clustering completed successfully.
```

CUSTOMER SEGMENT INTERPRETATION

After clustering, descriptive statistics are calculated for each cluster.

For every cluster, the following can be analyzed:

- Number of customers
- Average tenure
- Average monthly charges
- Average total charges
- Contract distribution
- Service characteristics
- Churn percentage

The clusters should then be assigned meaningful descriptive names based on the actual observed characteristics.

For example:

Long-Term Loyal Customers

May contain customers with relatively long tenure and stable service relationships.

New or Short-Tenure Customers

May contain customers who have recently joined the service and may require stronger early-stage engagement.

High-Charge Customers

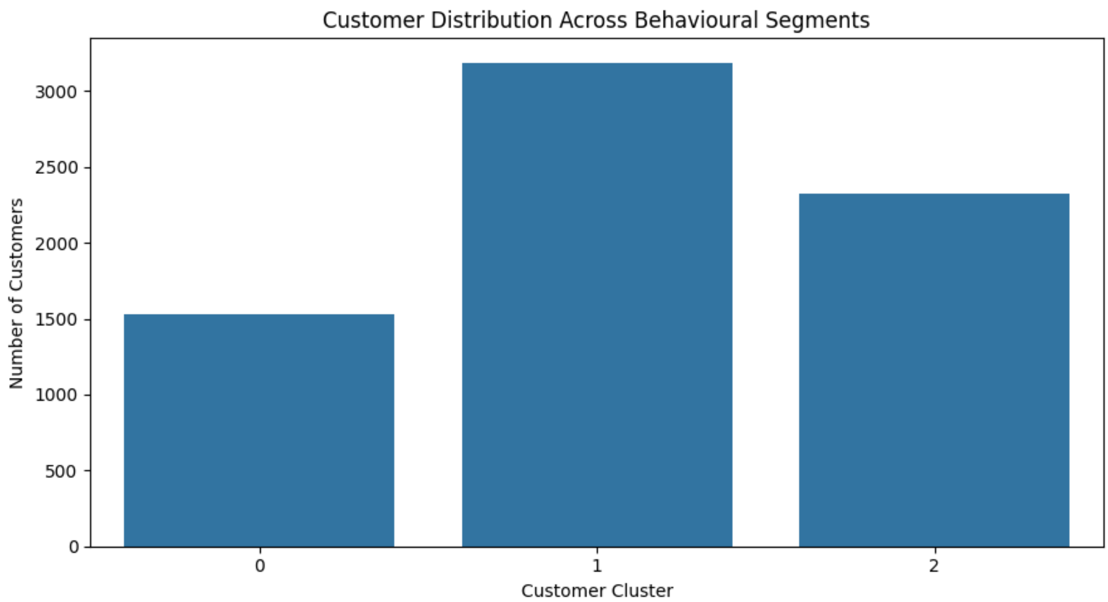
May represent customers with relatively high monthly charges and potentially greater price sensitivity.

High-Risk Behavioural Segment

May represent a segment with a comparatively high churn rate.

These labels should be finalized only after examining the actual clustering results.

Cluster	Customer Count
0	1526
1	3189
2	2328



CHURN RATE BY CUSTOMER SEGMENT

One of the important outputs of the project is the comparison of churn rates across clusters.

Cluster	Customer Count	Churned Customers	Churn Rate
Cluster 0	Actual	Actual	Actual
Cluster 1	Actual	Actual	Actual
Cluster 2	Actual	Actual	Actual
Cluster 3	Actual	Actual	Actual

This analysis connects unsupervised learning with the original churn problem.

A cluster containing a disproportionately high percentage of churned customers can be considered a priority segment for retention analysis.

```
================================================================================
CUSTOMER CLUSTER PROFILE
================================================================================
Average characteristics of each customer segment:
```

Cluster	tenure	MonthlyCharges	TotalCharges	SeniorCitizen	Churn
0	30.55	21.08	668.10	0.03	0.07
1	15.30	67.86	1020.27	0.20	0.44
2	56.95	89.15	5068.03	0.20	0.15

```
...
=======================================================================
CATEGORICAL CHARACTERISTICS OF CLUSTERS
=======================================================================

Contract Distribution Across Clusters (%):
```

Cluster	Month-to-month	One year	Two year
0	34.34	23.85	41.81
1	87.02	10.50	2.48
2	24.74	33.25	42.01

```
InternetService Distribution Across Clusters (%):
```

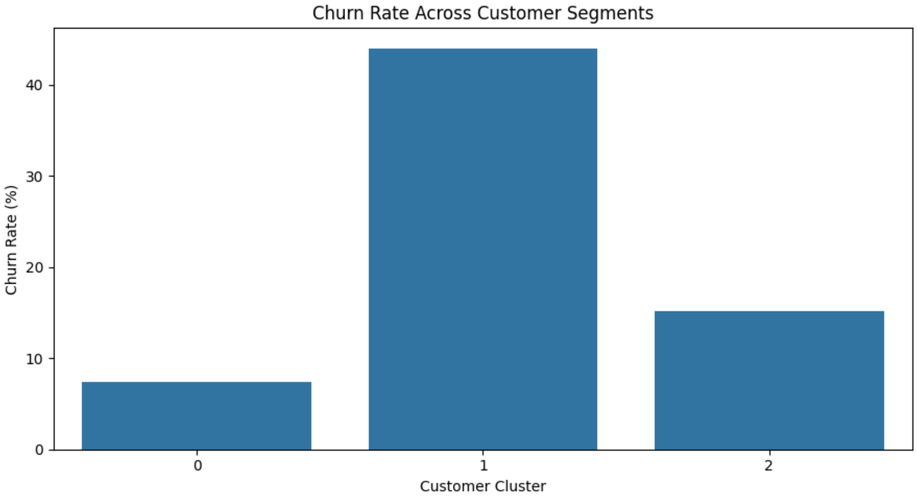
Cluster	DSL	Fiber optic	No
0	0.00	0.00	100.0
1	47.66	52.34	0.0
2	38.70	61.30	0.0

```
PaymentMethod Distribution Across Clusters (%):
```

Cluster	Bank transfer (automatic)	Credit card (automatic)	Electronic check	Mailed check
0	21.76	21.69	7.99	48.56
1	14.46	14.77	49.33	21.45
2	32.26	30.93	28.78	8.03

```
...
================================================================================
CHURN RATE BY CUSTOMER SEGMENT
================================================================================
```

Cluster	Customers	Churned	Churn_Rate
0	1526	113	7.40
1	3189	1404	44.03
2	2328	352	15.12



PCA-BASED CUSTOMER SEGMENT VISUALIZATION

After preprocessing and feature transformation, the dataset may contain a large number of dimensions, especially after categorical encoding.

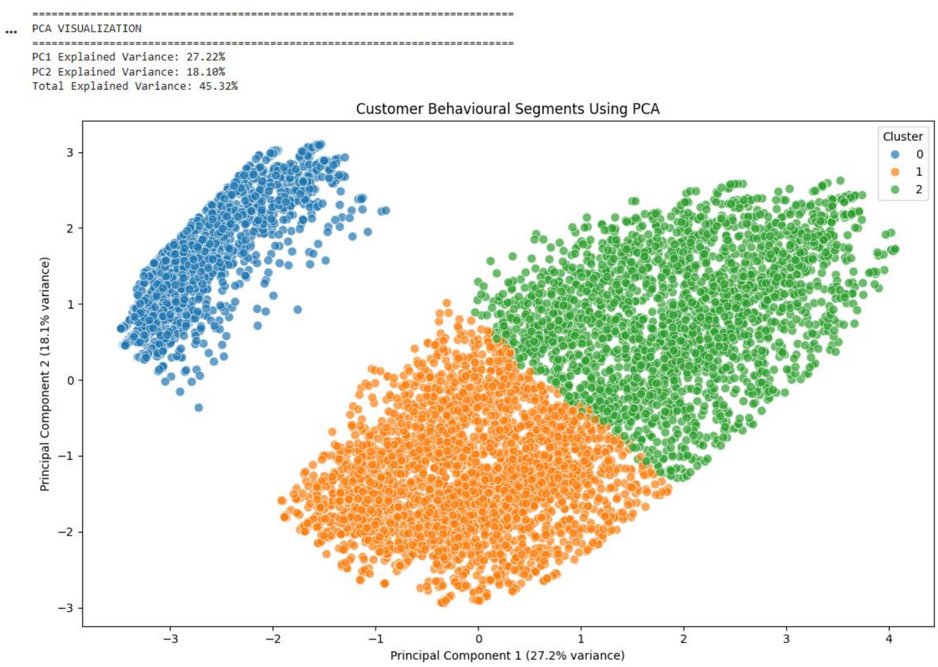
Principal Component Analysis can be used to transform the high-dimensional feature space into a smaller number of components.

The first two principal components can then be plotted to visually examine customer clusters.

The visualization should contain:

- PC1 on the horizontal axis
- PC2 on the vertical axis
- Cluster membership represented using different visual markers/colours

The PCA plot provides an intuitive representation of the degree to which the identified customer groups are separated in the transformed feature space.



INTEGRATED CUSTOMER RISK AND SEGMENT ANALYSIS

The key proposed feature of the system is the combination of classification and clustering results.

For each customer, the system can provide:

Customer Profile

↓

Churn Probability

↓

Risk Category

↓

Behavioural Segment

↓

Recommended Retention Strategy

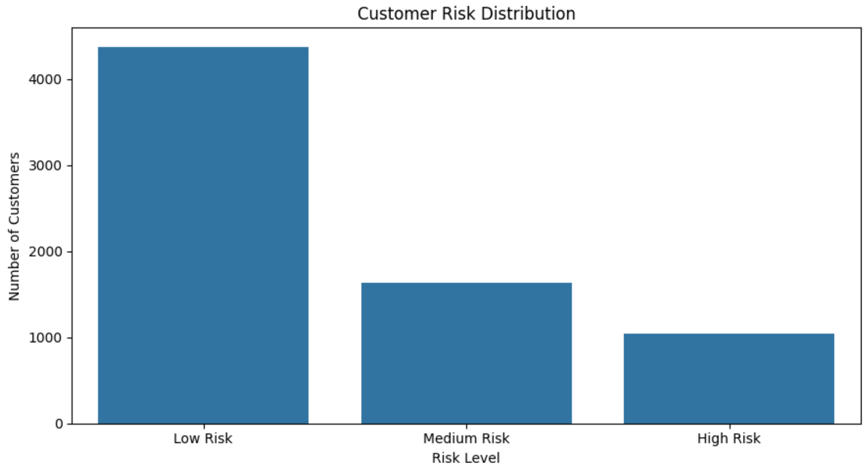
Example output:

Customer	**Risk**	**Segment**	**Recommended Action**
Customer A	High	Short-Tenure	Early engagement campaign
Customer B	High	High-Charge	Personalized pricing plan
Customer C	Medium	Moderate-Engagement	Loyalty incentive
Customer D	Low	Long-Term	Maintain relationship

This integrated approach allows the company to prioritize customers instead of applying identical retention strategies to everyone.

```
===========================================================================
CUSTOMER RISK ANALYSIS
===========================================================================
Risk thresholds used:
Low Risk    : Probability < 0.30
Medium Risk : 0.30 <= Probability < 0.60
High Risk   : Probability >= 0.60

Risk Distribution:
```

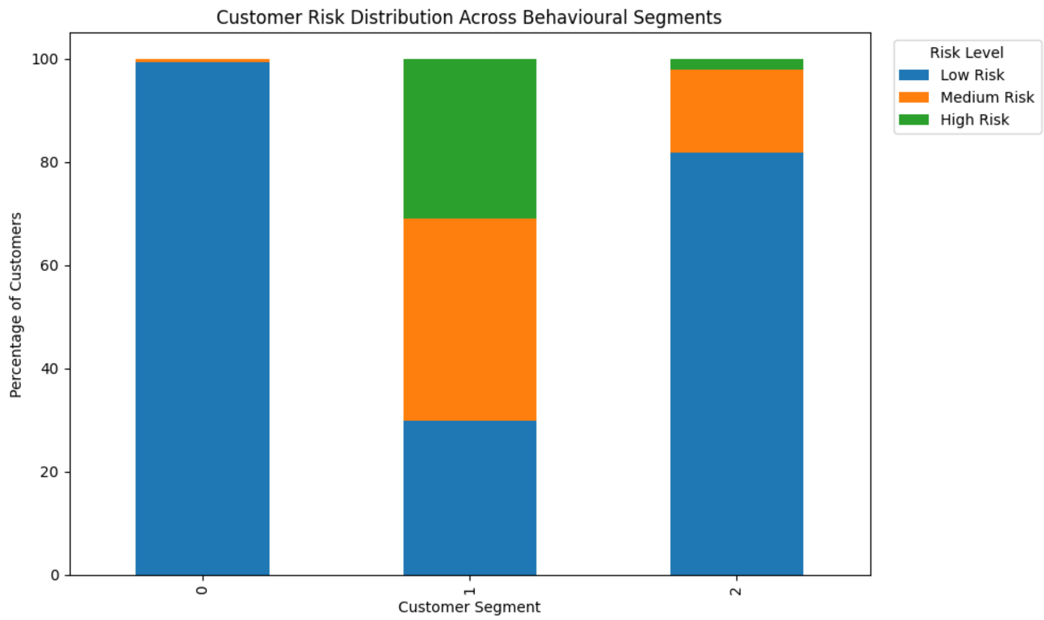
Risk_Level	Customers
Low Risk	4374
Medium Risk	1630
High Risk	1039




```
===========================================================================
RISK DISTRIBUTION ACROSS BEHAVIOURAL SEGMENTS
===========================================================================
```

Risk_Level Cluster	Low Risk	Medium Risk	High Risk
0	99.34	0.66	0.00
1	29.85	39.20	30.95
2	81.87	15.89	2.23

```
<Figure size 1000x600 with 0 Axes>
```




```
Total High-Risk Customers: 1039
Percentage of Customers Classified as High Risk: 14.75 %

High-Risk Customers by Cluster:
```

Cluster	High-Risk Customers
1	987
2	52

```
===========================================================================
CUSTOMER PRIORITY IDENTIFICATION
===========================================================================
```

Priority	Customers
Low Priority	4374
Medium Priority	2669

RETENTION RECOMMENDATIONS

The final recommendations should be derived from the actual characteristics of the identified customer segments.

High-Risk New Customers

Recommended actions:

- Strengthen onboarding.
- Provide early engagement support.
- Offer introductory incentives.

High-Charge Customers

Recommended actions:

- Provide personalized service plans.
- Introduce suitable bundled packages.
- Review pricing-related concerns.

Low-Engagement Customers

Recommended actions:

- Send targeted engagement campaigns.
- Promote relevant services.
- Provide personalized service assistance.

Loyal Customers

Recommended actions:

- Provide loyalty benefits.
- Offer personalized upgrades.
- Maintain consistent customer engagement.

These recommendations are decision-support suggestions rather than predictions generated directly by the ML models.

```
================================================================================
SEGMENT-LEVEL RETENTION STRATEGIES
================================================================================
```

	Cluster	Customers	Average Tenure	Average Monthly Charges	Churn Rate (%)	Average Churn Probability (%)	High-Risk Customers	Recommended Strategy
0	0	1526	30.55	21.08	7.40	7.23	0	Maintain customer engagement, loyalty programs...
1	1	3189	15.30	67.86	44.03	44.25	987	Targeted engagement, service improvements, loy...
2	2	2328	56.95	89.15	15.12	15.07	52	Maintain customer engagement, loyalty programs...

```
...
==========================================================================
RETENTION RECOMMENDATION ENGINE
==========================================================================
Sample Retention Recommendations:
```

	customerID	Cluster	Churn_Probability	Risk_Level	Priority	Retention_Recommendation
1976	9497-QCMMS	1	0.856422	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
3380	5178-LMXOP	1	0.855456	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
4800	9300-AGZNL	1	0.855336	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
3749	4424-TKOPW	1	0.852308	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
6368	2720-WGKHP	1	0.851503	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
5989	5567-WSELE	1	0.849250	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
1410	7024-OHCCK	1	0.848766	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
3159	5150-ITWWB	1	0.847357	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
2208	7216-EWTRS	1	0.845752	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
997	1374-DMZUI	1	0.841909	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
301	8098-LLAZX	1	0.837986	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
3209	8149-RSOUN	1	0.833623	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
6866	0295-PPHDO	1	0.832328	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
2577	4910-GMJOT	1	0.831578	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
2631	6861-XWTWQ	1	0.829096	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
1148	7851-WZEKY	1	0.826158	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
5282	3049-SOLAY	1	0.825113	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
6482	5419-JPRRN	1	0.823749	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
969	3158-MOERK	1	0.823000	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...
2745	4826-XTSOH	1	0.822643	High Risk	Medium Priority	Immediate proactive retention contact | Offer ...

IMPLEMENTATION

The implementation is developed using Python and standard machine learning libraries.

Major Libraries

Python

Pandas

NumPy

Scikit-learn

Matplotlib

Seaborn

If an interactive application is developed:

Streamlit

can be used for the interface.

Implementation Components

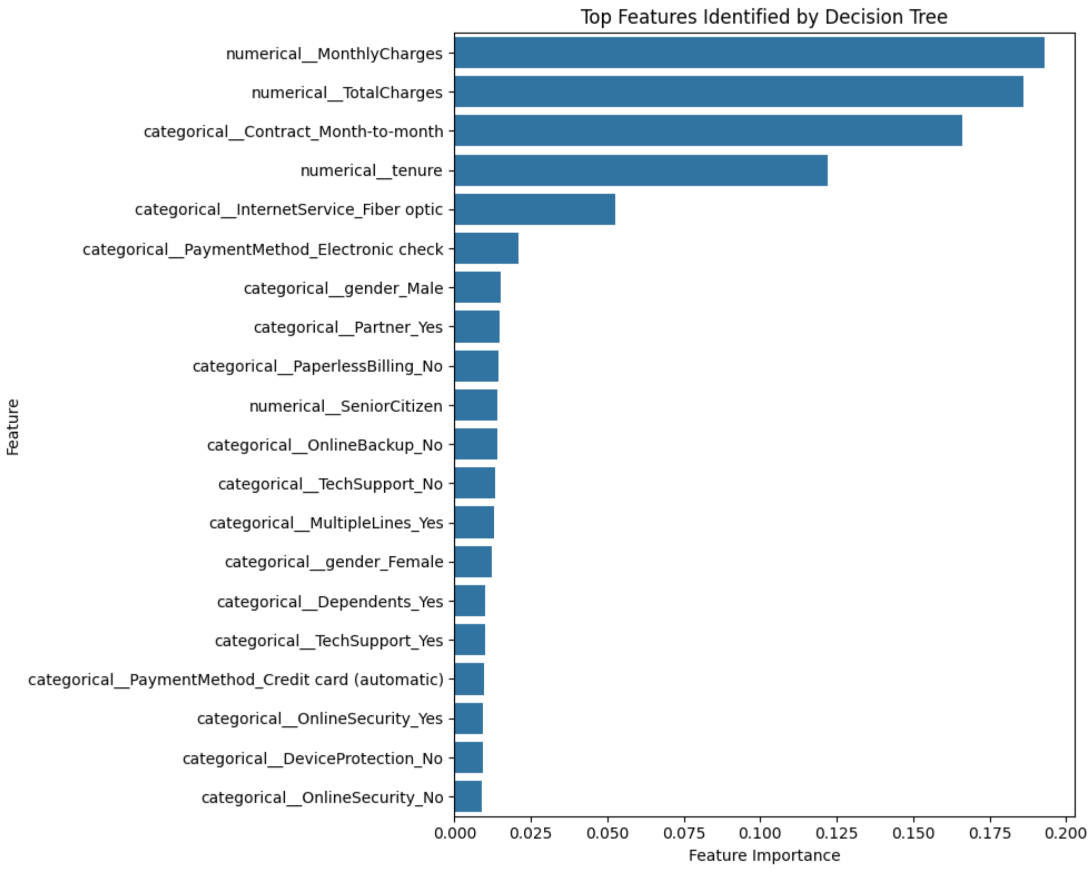
The implementation contains separate logical components for:

- Dataset loading

- Data validation
- Preprocessing
- Exploratory analysis
- Feature transformation
- Model training
- Model evaluation
- Hyperparameter experimentation
- K-Means clustering
- Cluster evaluation
- PCA
- Visualization
- Risk prediction
- Retention recommendations

```
===========================================================================
DECISION TREE FEATURE IMPORTANCE
===========================================================================
```

	Feature	Importance
2	numerical__MonthlyCharges	0.193076
3	numerical__TotalCharges	0.185979
36	categorical__Contract_Month-to-month	0.166082
1	numerical__tenure	0.121907
16	categorical__InternetService_Fiber optic	0.052756
43	categorical__PaymentMethod_Electronic check	0.020769
5	categorical__gender_Male	0.015292
7	categorical__Partner_Yes	0.014840
39	categorical__PaperlessBilling_No	0.014444
0	numerical__SeniorCitizen	0.014059
21	categorical__OnlineBackup_No	0.014009
27	categorical__TechSupport_No	0.013149
14	categorical__MultipleLines_Yes	0.012877
4	categorical__gender_Female	0.012060
9	categorical__Dependents_Yes	0.009939
29	categorical__TechSupport_Yes	0.009857
42	categorical__PaymentMethod_Credit card (automa...	0.009685
20	categorical__OnlineSecurity_Yes	0.009306
24	categorical__DeviceProtection_No	0.009272
18	categorical__OnlineSecurity_No	0.008971




```
TOP HIGH-RISK CUSTOMER TABLE
==========================================================================
```

	customerID	tenure	Contract	InternetService	MonthlyCharges	TotalCharges	Churn	Cluster	Churn_Probability	Risk_Level	Segment_Churn_Rate	Priority	Retention_Recommendation
1976	9497-QCMMS	1	Month-to-month	Fiber optic	93.55	93.55	1	1	0.856422	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
3380	5178-LMXOP	1	Month-to-month	Fiber optic	95.10	95.10	1	1	0.855456	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
4800	9300-AGZNL	1	Month-to-month	Fiber optic	94.00	94.00	1	1	0.855336	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
3749	4424-TKOPW	2	Month-to-month	Fiber optic	93.85	196.75	1	1	0.852308	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
6368	2720-WGKHP	2	Month-to-month	Fiber optic	94.00	181.70	1	1	0.851503	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
5989	5567-WSELE	3	Month-to-month	Fiber optic	94.60	279.55	1	1	0.849250	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
1410	7024-OHCCK	2	Month-to-month	Fiber optic	93.85	170.85	1	1	0.848766	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
3159	5150-ITWWB	3	Month-to-month	Fiber optic	94.85	335.75	0	1	0.847357	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
2208	7216-EWTRS	1	Month-to-month	Fiber optic	100.80	100.80	1	1	0.845752	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
997	1374-DMZUI	4	Month-to-month	Fiber optic	94.30	424.45	1	1	0.841909	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
301	8098-LLAZX	4	Month-to-month	Fiber optic	95.45	396.10	1	1	0.837986	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
3209	8149-RSOUN	1	Month-to-month	Fiber optic	93.85	93.85	1	1	0.833623	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
6866	0295-PPHDO	1	Month-to-month	Fiber optic	95.45	95.45	1	1	0.832328	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
2577	4910-GMJOT	1	Month-to-month	Fiber optic	94.60	94.60	1	1	0.831578	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
2631	6861-XWTWQ	7	Month-to-month	Fiber optic	99.25	665.45	1	1	0.829096	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
1148	7851-WZEKY	2	Month-to-month	Fiber optic	95.15	196.90	1	1	0.826158	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
5282	3049-SOLAY	3	Month-to-month	Fiber optic	95.20	292.85	1	1	0.825113	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
6482	5419-JPRRN	1	Month-to-month	Fiber optic	101.45	101.45	1	1	0.823749	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
969	3158-MOERK	2	Month-to-month	Fiber optic	96.00	174.80	1	1	0.823000	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
2745	4826-XTSOH	1	Month-to-month	Fiber optic	86.05	86.05	1	1	0.822643	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
6365	8884-ADFVN	7	Month-to-month	Fiber optic	101.95	700.85	1	1	0.822565	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
1600	3068-OMWZA	1	Month-to-month	Fiber optic	88.80	88.80	1	1	0.819828	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
1026	4822-RVYBB	8	Month-to-month	Fiber optic	100.60	819.40	1	1	0.819712	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
4585	1069-XAIEM	1	Month-to-month	Fiber optic	85.05	85.05	1	1	0.819141	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...
3346	2545-EBUPK	2	Month-to-month	Fiber optic	84.05	186.05	0	1	0.817977	High Risk	44.03	Medium Priority	Immediate proactive retention contact | Offer ...

```
==========================================================================
SAVING MACHINE LEARNING MODELS
==========================================================================
✓ Best churn model saved.
✓ All classification models saved.
✓ Cluster preprocessing pipeline saved.
✓ K-Means model saved.
✓ PCA model saved.

PROJECT SUMMARY:
```

	Total Customers	Total Features	Overall Churn Rate (%)	Best Classification Model	Best Accuracy	Best Precision	Best Recall	Best F1-Score	Best ROC-AUC	Optimal K	Silhouette Score	Low Risk Customers	Medium Risk Customers	High Risk Customers
0	7043	19	26.54	Logistic Regression	0.8055	0.6572	0.5588	0.604	0.8419	3	0.2388	4374	1630	1039

```
================================================================================
FINAL PROJECT RESULTS
================================================================================

DATASET
--------------------------------------------------------------------------------
Total Customers: 7043
Total Input Features: 19
Overall Churn Rate: 26.54 %

CLASSIFICATION
--------------------------------------------------------------------------------
Models Evaluated:
  ✓ Logistic Regression
  ✓ Naive Bayes
  ✓ Decision Tree
  ✓ Support Vector Machine

Best Model: Logistic Regression
Accuracy: 0.8055
Precision: 0.6572
Recall: 0.5588
F1-Score: 0.604
ROC-AUC: 0.8419

CLUSTERING
--------------------------------------------------------------------------------
Optimal Number of Clusters: 3
Silhouette Score: 0.2388

RISK ANALYSIS
--------------------------------------------------------------------------------
Low Risk Customers: 4374
Medium Risk Customers: 1630
High Risk Customers: 1039
```

```
HIGH-RISK SEGMENTS
--------------------------------------------------------------------------------
```

Cluster	Customers	Average_Churn_Probability	Actual_Churn_Rate	High_Risk_Count
0	1526	7.23	7.40	0
1	3189	44.25	44.03	987
2	2328	15.07	15.12	52

```
================================================================================
CREATING PROJECT OUTPUT ZIP
================================================================================

ZIP file created successfully:
/content/customer_churn_project_complete.zip
```

IMPORTANT IMPLEMENTATION OUTPUTS

The following outputs should be generated and captured as screenshots for the report:

- Dataset preview
- Dataset dimensions
- Missing-value analysis
- Preprocessing output
- Churn distribution

- Important EDA visualizations
- Model training output
- Model performance comparison
- Confusion matrices
- ROC curves, if implemented
- Hyperparameter comparison
- Elbow Method graph
- Silhouette score comparison
- K-Means cluster distribution
- PCA cluster visualization
- Cluster characteristics
- Churn rate by cluster
- Customer risk prediction
- Final application/dashboard

These screenshots provide evidence that the implementation was actually executed.

```
================================================================================
... GENERATED PROJECT FILES
================================================================================
customer_churn_project/
  models/
    kmeans_model.pkl
    pca_model.pkl
    all_classification_models.pkl
    best_churn_model.pkl
    cluster_preprocessor.pkl
  results/
    elbow_results.csv
    final_model_comparison.csv
    segment_summary.csv
    cluster_counts.csv
    project_summary.json
    cluster_paymentmethod_distribution.csv
    cluster_churn.csv
    cluster_profile.csv
    high_risk_by_cluster.csv
    customer_risk_analysis.csv
    cluster_contract_distribution.csv
    high_risk_cluster_summary.csv
    top_100_risk_customers.csv
    processed_customer_data.csv
    risk_distribution_by_cluster.csv
    pca_coordinates.csv
    silhouette_results.csv
    tuned_decision_tree_results.csv
    model_comparison.csv
    project_summary.csv
    top_decision_tree_features.csv
    cluster_internetservice_distribution.csv
  figures/
    17_customer_risk_distribution.png
    09_confusion_matrices.png
    06_monthly_charges_vs_churn.png
    07_tenure_monthlycharges.png
    13_silhouette_scores.png
    14_cluster_distribution.png
    19_decision_tree_feature_importance.png
    16_pca_customer_segments.png
    12_elbow_method.png
    01_churn_distribution.png
    03_internet_service_vs_churn.png
    15_cluster_churn_rate.png
    11_roc_curve_comparison.png
    05_tenure_vs_churn.png
    02_contract_vs_churn.png
    18_risk_by_cluster.png
    08_correlation_heatmap.png
    04_payment_method_vs_churn.png
    10_model_comparison.png
```

```
================================================================================
IMPLEMENTATION COMPLETED SUCCESSFULLY
================================================================================

Your project now contains:

✓ Dataset loading and understanding
✓ Data cleaning and preprocessing
✓ Missing-value handling
✓ Duplicate handling
✓ Exploratory Data Analysis
✓ Feature transformation
✓ Train-test split
✓ Logistic Regression
✓ Naive Bayes
✓ Decision Tree
✓ Support Vector Machine
✓ Accuracy
✓ Precision
✓ Recall
✓ F1-Score
✓ ROC-AUC
✓ Classification reports
✓ Confusion matrices
✓ ROC curves
✓ Model comparison
✓ Best model selection
✓ Hyperparameter experimentation
✓ K-Means clustering
✓ Elbow Method
✓ Silhouette Score
✓ Optimal K selection
✓ Cluster profiling
✓ Churn rate by cluster
✓ PCA visualization
✓ Churn probability
✓ Customer risk classification
✓ High-risk customer identification
✓ Priority identification
✓ Retention recommendations
✓ Segment-level retention strategies
✓ Feature importance
✓ Saved ML models
✓ Saved CSV/JSON results
✓ Complete project ZIP

The generated ZIP contains the important files needed later
for your report and potential Google AI Studio web application.
```

IMPLEMENTATION TESTING

The completed implementation should be tested from beginning to end to ensure that all components operate correctly.

Test Case	**Testing Objective**	**Expected Result**
Dataset Loading	Verify dataset can be loaded	Dataset displayed successfully
Data Preprocessing	Verify missing values and categorical data are handled	Clean feature matrix generated
Model Training	Verify all selected models train successfully	Models trained without errors
Prediction	Verify unseen customer data can be classified	Churn prediction generated

Evaluation	Verify metrics are calculated	Performance metrics displayed
K-Means	Verify clustering works	Customer clusters generated
PCA	Verify dimensionality reduction	2D visualization generated
Risk Analysis	Verify probability-based risk assignment	Risk category generated
Recommendation	Verify segment-based recommendation	Retention action displayed
Complete Execution	Verify complete workflow	Project executes from beginning to end

The assignment specifically states that the implementation should be executable from beginning to end without errors.

RESULTS AND DISCUSSION

The final results should discuss both classification and clustering outcomes.

The classification results should identify which supervised learning model provides the most suitable performance according to the selected evaluation criteria.

The clustering results should explain the characteristics of the identified customer groups.

The relationship between clusters and churn should then be examined to identify customer segments requiring greater attention.

The most important result of the project is therefore not simply the classification accuracy. It is the ability to transform customer data into actionable information:

FINAL OBSERVATIONS

The following observations should be finalized after inserting the actual experimental results:

- Customer churn is influenced by multiple demographic, service, contractual and billing-related characteristics rather than a single variable.
- Data preprocessing plays an important role because the dataset contains categorical variables, numerical attributes and potentially inconsistent values.
- Different supervised learning algorithms produce different levels of classification performance.
- Accuracy alone may not be sufficient for evaluating a churn prediction system because the ability to identify actual churners is particularly important.
- K-Means clustering provides an additional perspective by identifying groups of customers with similar characteristics.
- PCA provides a useful visual representation of customer clusters in reduced-dimensional space.
- Some customer segments may demonstrate higher churn rates than others.
- Combining churn probability with customer segmentation provides more useful information for targeted retention planning.
- The final system can help organizations prioritize retention efforts toward customers and customer groups with greater churn risk.

CONCLUSION

The proposed Intelligent Customer Churn Risk Prediction and Behavioural Segmentation System demonstrates a complete machine learning workflow for analyzing customer attrition.

The system performs data preprocessing and exploratory analysis before applying multiple supervised learning algorithms for churn prediction. The implemented models are evaluated using accuracy, precision, recall, F1-score and confusion matrices, allowing their performance to be compared objectively.

K-Means clustering is then applied to discover customer groups based on behavioural and service-related characteristics. The Elbow Method and Silhouette Score can be used to support the selection of an appropriate number of clusters, while PCA provides a visual representation of the resulting segments.

The integration of churn prediction and customer segmentation provides a more practical approach than using classification alone. It allows the system to identify not only customers who may churn, but also the type of customer they represent and the broader customer segments requiring attention.

The final outcome is a machine-learning-based decision-support framework that can assist service organizations in identifying high-risk customers, understanding customer behaviour and developing more targeted retention strategies.

FUTURE SCOPE

Real-Time Churn Monitoring

- Integrate continuously updated customer activity and service data.
- Generate dynamic churn-risk estimates as customer behaviour changes.

Explainable Machine Learning

- Integrate techniques such as SHAP or LIME.
- Explain the major factors contributing to individual churn predictions.

Advanced Segmentation

- Compare K-Means with alternative clustering techniques.
- Investigate whether different algorithms reveal additional customer behaviour patterns.

Cost-Aware Retention

- Incorporate the financial cost of retention campaigns.
- Prioritize interventions based on expected business benefit.

Automated Retention Optimization

- Develop personalized intervention recommendations.
- Evaluate the effectiveness of different retention strategies over time.

REFERENCES

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Individual Contribution

I contributed to the development of the Intelligent Customer Churn Risk Prediction and Behavioural Segmentation System by actively working on the major stages of the project, from understanding the problem and dataset to implementing machine learning models and analysing the results.

My primary contribution was in dataset understanding and preprocessing. I worked with the Telco Customer Churn dataset and examined its attributes, data types, missing values, and target variable. I helped identify irrelevant attributes such as customerID, converted TotalCharges into a suitable numerical format, handled missing values, encoded categorical variables, and prepared the data for machine learning. I also ensured that the preprocessing steps were applied consistently to the training and testing data.

I also contributed to the Exploratory Data Analysis (EDA) stage. I analysed the distribution of churn and studied customer-related characteristics such as tenure, monthly charges, contract type, internet service, and other service-related attributes. The visualizations generated during this stage helped in understanding customer behaviour and identifying factors that could be associated with churn.

A major part of my work involved the implementation and evaluation of supervised machine learning models. I worked with Logistic Regression, Gaussian Naive Bayes, Support Vector Machine (SVM), and Decision Tree models for predicting whether a customer is likely to churn. I evaluated the models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Based on the experimental results, Logistic Regression achieved the best overall F1-score of 0.6040 and ROC-AUC of 0.8419, making it the selected classification model for the system.

I further contributed to Decision Tree hyperparameter experimentation using GridSearchCV. The best configuration obtained a maximum depth of 5 and was evaluated on the test set to compare its performance with the other classification models. This helped demonstrate that model selection should be based on multiple evaluation metrics rather than accuracy alone.

I also worked on the customer segmentation component using K-Means clustering. I experimented with different numbers of clusters using the Elbow Method and Silhouette Score and selected three meaningful customer groups. I analysed the clusters based on customer count, average tenure, monthly charges, and churn rate. The analysis showed that the short-tenure, higher-charge customer segment had the highest churn rate and therefore represents an important group for targeted retention strategies.

Another contribution was the use of PCA for visualization of the customer segments. PCA helped reduce the dimensionality of the prepared customer data into two principal components so that the clustering structure could be visually interpreted.

I also contributed to the integration of prediction and segmentation results by developing the concept of customer churn-risk levels such as Low, Medium, and High Risk. The system combines predicted churn probability with behavioural segment information to identify customers who may require greater retention attention. I helped formulate suitable retention recommendations based on characteristics such as tenure, contract type, monthly charges, and customer service usage.

Finally, I contributed to the overall project documentation and presentation of results, including preparing workflow descriptions, pseudocode, methodology, model comparison, cluster interpretation, visualizations, observations, conclusion, and report content. I also worked on organizing the implementation so that the complete workflow can be executed in Google Colab from data loading and preprocessing through model training, clustering, visualization, evaluation, and generation of final results.

Overall, my contribution covered the core technical and analytical stages of the project, particularly data preprocessing, exploratory analysis, supervised machine learning, model evaluation and tuning, K-Means segmentation, PCA visualization, churn-risk analysis, retention recommendations, and documentation of the complete system.